

Auto-Assign Advisor -- Implementation Plan

Overview

Automatically assign the best-available advisor to a lead when the car checks in, using the existing PIS scoring engine. No manual "Assign" button needed at this stage.

Trigger

Event	Condition
Lead stage changes to <code>checkin</code> (Car In)	<code>lead_stage = 'checkin' OR lead_status = 'car_in'</code>
Source	PATCH <code>/api/company/[companyId]/sales/leads/[id]</code> with <code>status: "car_in"</code>

The auto-assignment fires **once** when the car-in event occurs. It does not re-trigger if the lead is already assigned.

Selection Criteria (Using PIS Engine)

Step 1: Get Eligible Advisors

Filter	Rule	Source
Department	Only "Service Center Sales Department" employees	<code>employees.department</code>
Company	Same company as the lead	<code>employees.company_id = lead.company_id</code>
Active	<code>is_active = true</code>	<code>employees.is_active</code>
Not Locked	Not currently handling another lead (within 120 min lock window)	<code>pis_lead_queue.status != 'locked'</code>

Note: No branch filter. Car check-in can happen at any branch (including recovery drop-off at a different branch).

Step 2: Rank by PIS Composite Score

Use the existing PIS weighted scoring:

Metric	Weight	Description
Conversion Rate	25%	Leads converted to closed_won / total leads
Gross Profit %	20%	(Revenue - Cost) / Revenue

Metric	Weight	Description
Revenue	15%	Normalized total revenue
SLA Compliance	15%	Estimate approval/invoice rate
Customer Satisfaction	10%	Call completion / lead health
Call Quality	10%	Completed calls %
FOC Penalty	-5%	Free-of-charge invoice rate

Composite Score = weighted sum of normalized metrics (0-100 scale)

Advisors are sorted **descending** by composite score. The highest-scoring available advisor is offered the lead first.

Step 3: Determine Tier & Accept Window

Tier	Score Range	Accept Window
Tier 1 (ELITE)	>= 80	10 minutes
Tier 2 (STANDARD)	>= 50	7 minutes
Tier 3 (LOW)	< 50	5 minutes

Step 4: Offer Lead to Advisor

1. Create `pis_lead_queue` entry with `status = 'offered'`
2. Set `offered_to_user_id`, `offered_at`, `locked_until` (now + accept window)
3. Log `pis_lead_queue_history` with `action = 'offered'`
4. Log `lead_events` with `event_type = 'advisor_auto_assigned'`
5. Send push notification to advisor (Firebase)

Step 5: Wait for Acceptance

- Advisor sees the lead in their portal / mobile app
- Advisor clicks "Accept" -> `status = 'accepted'`, `accepted_at = now()`, lock for 120 min
- Log history: `action = 'accepted'`

Step 6: Cascade on Timeout

If advisor does not accept within the accept window:

1. Apply **5-point penalty** to timed-out advisor
2. Log history: `action = 'timeout'`, `penalty_points = 5`
3. Increment `cascade_count`
4. Check if `cascade_count >= 5` (max cascades)
 - **Yes:** Set `status = 'manual'`, notify branch manager
 - **No:** Offer to next best available advisor (repeat from Step 3)

Implementation Steps

Phase 1: Backend Hook (Trigger)

File: `packages/ai-core/src/crm/leads/repository.ts` (or new service file)

1. Create function `triggerAdvisorAutoAssign(companyId, leadId)`
2. Call this function inside `updateLeadPartial()` when:
 - `leadStatus` changes to `car_in`, OR
 - `leadStage` changes to `checkin`
3. Guard: Skip if lead already has `agent_employee_id` set (already assigned)
4. Guard: Skip if `pis_lead_queue` already has an entry for this lead

Phase 2: Reuse PIS Engine

File: `packages/ai-core/src/pis/leadDistribution/engine.ts`

The existing `routeLead(companyId, leadId, pipelineValue)` function already does everything needed:

- Fetches advisors sorted by composite score
- Filters out locked advisors
- Creates queue entry with offer
- Handles tier-based accept windows

We just need to call `routeLead()` from the car-in trigger.

Pipeline value: Calculate from the estimate linked to the lead (if exists), otherwise use 0.

Phase 3: Auto-Check Expired Offers

File: `packages/ai-core/src/pis/leadDistribution/engine.ts`

The existing `checkExpiredOffers(companyId)` function handles cascade logic. It needs to run periodically:

Option A: Call it on every leads API request (current approach with `releaseExpiredAssignments`) **Option B (Recommended):** Set up a cron job / interval that runs every 1-2 minutes

Phase 4: Update Lead on Acceptance

When advisor accepts (POST `/pis/lead-distribution/accept`):

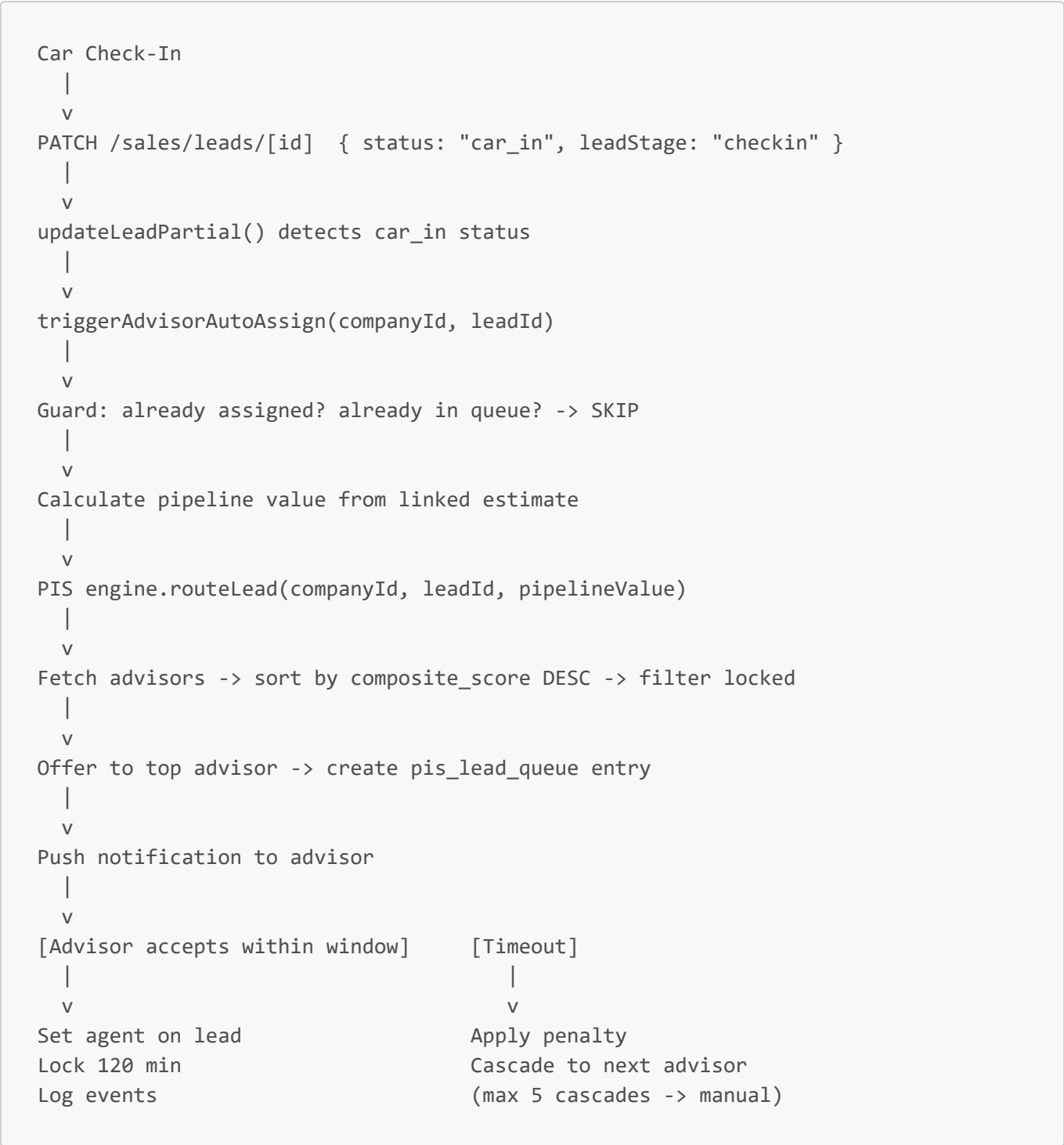
1. Set `leads.agent_employee_id` = accepted advisor's employee ID
2. Set `leads.assigned_user_id` = accepted advisor's user ID
3. Set `leads.assigned_at` = `now()`
4. Log `lead_events` with `event_type` = `'advisor_accepted'`

Phase 5: Notifications

1. **On offer:** Push notification to advisor via Firebase
 - Title: "New Lead Assigned"
 - Body: "Customer {name} - {car} checked in. Accept within {window} minutes."
2. **On timeout cascade:** Push notification to next advisor

- 3. **On manual escalation:** Push notification to branch manager
- 4. **On acceptance:** Push notification to front-desk / check-in agent confirming advisor assigned

Data Flow



Files to Modify

File	Change
packages/ai-core/src/crm/leads/repository.ts	Add auto-assign trigger in updateLeadPartial()

File	Change
packages/ai-core/src/pis/leadDistribution/engine.ts	Reuse routeLead() -- no changes needed
apps/web/app/api/company/[companyId]/pis/lead-distribution/accept/route.ts	Add lead field updates on acceptance
packages/ai-core/src/pis/advisorRepository.ts	No changes -- scoring already works
packages/ui/src/components/leads/LeadsTable.tsx	Remove manual Assign button (after auto-assign is live)

New Files

File	Purpose
packages/ai-core/src/pis/leadDistribution/autoAssign.ts	Auto-assign service: trigger, guards, pipeline value calculation

Edge Cases

Scenario	Handling
No advisors available (all locked)	Set queue status to pending, retry when checkExpiredOffers runs
No advisors in system	Skip auto-assign, log warning, keep lead unassigned
Lead already assigned	Skip (guard check on agent_employee_id)
Lead already in queue	Skip (guard check on existing pis_lead_queue entry)
Advisor goes offline after offer	Timeout will cascade automatically
Multiple car check-ins for same lead	Guard prevents duplicate queue entries
VIP customer	Future: add priority flag to skip workload check, assign to top performer

Testing Checklist

#	Test	Expected
1	Car check-in with available advisors	Top-scored advisor gets offer, queue entry created
2	Advisor accepts within window	Lead gets agent_employee_id, queue status = accepted
3	Advisor times out	Penalty applied, cascades to next advisor
4	5 timeouts	Queue status = manual, branch manager notified

#	Test	Expected
5	All advisors locked	Queue stays pending , retries later
6	Lead already assigned	No queue entry created, skipped
7	Car check-in via recovery (different branch)	Works without branch filter
8	Push notification delivery	Advisor receives notification on mobile
9	Pipeline value from estimate	Correct value passed to queue
10	Multiple check-ins same lead	Only first one triggers assignment

Configuration (PIS Admin)

All settings are configurable via PIS Admin panel ([/company/\[companyId\]/pis/admin](/company/[companyId]/pis/admin)):

Setting	Default	Key
Elite min score	80	<code>tier_boundaries.elite_min_score</code>
Standard min score	50	<code>tier_boundaries.standard_min_score</code>
Tier 1 accept window	10 min	<code>lead_distribution.tier1_accept_window_min</code>
Tier 2 accept window	7 min	<code>lead_distribution.tier2_accept_window_min</code>
Tier 3 accept window	5 min	<code>lead_distribution.tier3_accept_window_min</code>
Lock duration	120 min	<code>lead_distribution.lock_duration_min</code>
Timeout penalty	5 points	<code>lead_distribution.no_call_penalty_points</code>
Max cascade attempts	5	<code>lead_distribution.max_cascade_attempts</code>
Score weights	See table above	<code>score_weights.*</code>